NeuroVision AI - Lean Canvas

Tumor Classification & Clinical Decision Support

Explainable Brain MRI

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1. PROBLEM

- **Critical radiologist shortage** globally: severe diagnostic delays in rural and tier-2/3 hospital networks.
- **Diagnostic fatigue & variability**: interpretation of subtle brain lesions (glioma vs meningioma) requires years of neuro-radiology specialization.
- **Black-box AI opacity**: existing deep learning models output raw class labels without visual justification, leading to clinical distrust.
- **Emergency triage bottlenecks**: acute brain scans often wait hours in PACS worklists before preliminary review.

EXISTING ALTERNATIVES

- Manual review by on-call radiologists (hours to days)
- General teleradiology

2. SOLUTION

- **4-Class Automated Brain MRI Classifier**: Accurately identifies Glioma, Meningioma, Pituitary tumor, and Normal tissue.
- **Transparent Explainable AI (Grad-CAM)**: Generates gradient-weighted visual saliency overlays highlighting exact lesion boundaries.
- **Automated PDF Radiology**

8. KEY METRICS

- **Classification Accuracy**: 96.38% overall on balanced multi-class test dataset (n=1,600).
- **Macro AUC**: 0.9880; per-class AUC 0.982 (Glioma), 0.984 (Meningioma), 0.995 (Pituitary), 0.991 (No Tumor).
- **Macro F1-Score**: 0.9632 (Weighted F1: 0.9639).
- **Inference Latency**: < 1.2s end-to-end (CPU/GPU optimized).
- **Grad-CAM Generation**: < 400ms

3. UNIQUE VALUE PROPOSITION

- **Rapid, Explainable Second-Opinion Radiology AI**: Upload any axial brain MRI and receive a 4-class differential diagnosis with visual anatomical proof in under 2 seconds.
- **Clinician-Centered Explainability**: Visual heatmaps isolate exact regions of concern, transforming AI from an inscrutable oracle into a trusted co-pilot.
- **Open, Accessible, and Honest**: Honest disclosure of diagnostic scope and limitations; designed for healthcare democratisation.

HIGH-LEVEL CONCEPT

- **“Spell-check for Brain MRIs”**: An instantaneous, explainable assistant.
- Standardized radiology pre-triage preliminary visual audit before ap

4. UNFAIR ADVANTAGE

- **Targeted 2-Stage Transfer Learning**: EfficientNetB0 backbone optimized with domain-adapted fine-tuning on brain MRI morphology.
- **Native Grad-CAM Integration**: Directly inspects activations from deepest feature layer (`top_conv`, 1280 channels) with no
- **Automated Vector PDF Engine**:

5. CHANNELS

- **Web Application**: Cloud-deployed interactive platform accessible on desktop and tablet.
- **Teaching Hospitals & Medical Colleges**: Radiology residency training and neuro-oncology workshops.
- **BharatCares & AICTE Network**: University deployment, academic showcase, and open-source
- **GitHub Repository**: Open-source code
- **Future**: DICOM/PACS plugin.

6. CUSTOMER SEGMENTS

- **Radiologists & Neuro-Oncologists**: Seeking rapid preliminary triage and verified diagnostic verification.
- **Medical Students & Radiology Trainees**: Educational learning tool for recognizing tumor morphology and MRI features.
- **Tier-2 & Rural Diagnostic Clinics**: Healthcare facilities lacking 24/7 on-site neuro-radiology specialists.
- **Clinical Research Labs**: Academic institutions conducting neuro-imaging and explainable AI studies.

EARLY ADOPTERS

- Medical college teaching hospitals and radiology residency programs.
- Tele-radiology triage units
- Academic biomedical imaging

7. COST STRUCTURE

- **Compute & Cloud Hosting**: GPU training instances (Google Colab / local RTX) and low-cost containerized web app hosting (Render / Docker).
- **Data Management & Storage**: Curated multi-sequence MRI dataset storage, preprocessing pipelines, and versioned artifact repository.
- **Model Retraining & Validation**: Continuous hyperparameter tuning, fine-tuning audits, and domain generalization across scanner types.
- **Open Source Maintenance**: Student-driven engineering, unit test maintenance, and documentation updates.

9. REVENUE STREAMS (HYPOTHESES TO VALIDATE)

- **Free Educational & Academic Tier**: Zero-cost open access for university students and medical trainees (aligned with UN SDG 3 & 9).
- **B2B Institutional Licensing**: Subscription-based decision-support integration for private diagnostic centers and hospital networks.
- **Tiered REST API Services**: Per-scan micro-billing for tele-radiology platforms requiring automated preliminary triage.
- **CSR & Healthcare Grants**: BharatCares, IBM SkillsBuild, and governmental digital health grants for rural diagnostic equity.

Lean Canvas format adapted from Ash Maurya (Creative Commons BY-SA 3.0). Performance metrics derived from NeuroVision AI test set evaluation (n=1,600, 96.38% accuracy, 0.988 AUC). All clinical business models represent academic hypotheses for translational deployment.